FTM 1 MODULE OBJECTIVES

Activity	Discipline	Title	OBJECTIVE CODE	Lecture Objective
DLA 1	BCHM; HCB	Membrane Structure and Transport: Integrated Basic Principles	SOM.MK.1.BPM1.1.FTM.3.HCB.0683	Differentiate between the two (2) major classes of membrane proteins.
			SOM.MK.1.BPM1.1.FTM.3.HCB.0684	List the two major categories of integral membrane proteins.
			SOM.MK.1.BPM1.1.FTM.3.HCB.0685	Describe the six major functions of the two (2) major categories of integral membrane proteins.
			SOM.MK.1.BPM1.1.FTM.3.HCB.0686	Discuss membrane permeability in relation to membrane composition, and hydrophilic and lipophilic compounds
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0010	Describe the structure, composition and significance of the fluid mosaic model of cell membrane
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0011	Review the lipid (cholesterol, phospholipids and glycolipids) composition of the plasma membrane. Discuss the relevance of cholesterol and lipid composition in regulating membrane fluidity
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0014	Differentiate between active and passive transport systems based on energy requirement and movement along the concentration gradient.
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0015	Discuss facilitated diffusion by glucose transporters (GLUT 1-5) highlighting their location and characteristics in relation to substrate affinity and insulin sensitivity.
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0017	Distinguish between primary active transport (Na+/K+ -ATPase) and secondary active transport (SGLT).
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0018	Describe the CFTR channel as an example of ABC transporters. Predict effects of CFTR mutations on lung and pancreas.
Lecture 1	HCB	Histology of the Cytoskeleton of the Cell	SOM.MK.1.BPM1.1.FTM.3.HCB.0709	List the three domains of cellular life.
			SOM.MK.1.BPM1.1.FTM.3.HCB.0710	List the three major types of protein filaments (cytoskeletal elements) that form the cytoskeleton.
			SOM.MK.1.BPM1.1.FTM.3.HCB.0711	State the function(s) of microtubules as a cytoskeletal element.
			SOM.MK.1.BPM1.1.FTM.3.HCB.0712	State the function(s) of intermediate filaments as a cytoskeletal element.
			SOM.MK.1.BPM1.1.FTM.3.HCB.0713	State the function(s) of microfilaments as a cytoskeletal element.
			SOM.MK.1.BPM1.1.FTM.3.HCB.0714	Describe the protein structure of microtubules as a cytoskeletal element.
			SOM.MK.1.BPM1.1.FTM.3.HCB.0715	Describe the protein structure of intermediate filaments as a cytoskeletal element.
			SOM.MK.1.BPM1.1.FTM.3.HCB.0716	Describe the protein structure of microfilaments as a cytoskeletal element.
			SOM.MK.1.BPM1.1.FTM.3.HCB.0717	Describe the assembly of microtubules.
			SOM.MK.1.BPM1.1.FTM.3.HCB.0718	Describe the assembly of intermediate filaments.
			SOM.MK.1.BPM1.1.FTM.3.HCB.0719	Describe the assembly of microfilaments.
			SOM.MK.1.BPM1.1.FTM.3.HCB.0720	Discuss three compounds that can affect microtubule function.
			SOM.MK.1.BPM1.1.FTM.3.HCB.0721	Discuss two compounds that can affect microfilament function.
			SOM.MK.1.BPM1.1.FTM.3.HCB.0722	State the function(s) of the centrosome.
			SOM.MK.1.BPM1.1.FTM.3.HCB.0723	Describe the structure of the centrosome.
			SOM.MK.1.BPM1.1.FTM.3.HCB.0724	Give the location of the basal body.
			SOM.MK.1.BPM1.1.FTM.3.HCB.0725	State the function(s) of the basal body.
			SOM.MK.1.BPM1.1.FTM.3.HCB.0726	Describe the structure of the basal body.
			SOM.MK.1.BPM1.1.FTM.3.HCB.0727	Differentiate between the following apical specializations based on function, tissues found and their microanatomical structures: primary cilia, motile cilia and flagella.
			SOM.MK.1.BPM1.1.FTM.3.HCB.0728	Differentiate between the following apical specializations based on function and their microanatomical structures: microvilli, stereocilia.
			SOM.MK.1.BPM1.1.FTM.3.HCB.0729	Describe the terminal web.
			SOM.MK.1.BPM1.1.FTM.3.HCB.0730	Describe the motor protein associated with microfilaments.
			SOM.MK.1.BPM1.1.FTM.3.HCB.0731	Describe the two motor proteins associated with microtubules.
			SOM.MK.1.BPM1.1.FTM.3.HCB.0732	Describe role of the cytoskeleton in intracellular transport.
			SOM.MK.1.BPM1.1.FTM.3.HCB.0733	Define actin polymerization.
			SOM.MK.1.BPM1.1.FTM.3.HCB.0734	Describe the three types of membrane protrusion structures.
			SOM.MK.1.BPM1.1.FTM.3.HCB.0735	Describe the role of actin polymerization in membrane protrusion.
			SOM.MK.1.BPM1.1.FTM.3.HCB.0736	Describe the role of actin polymerization in cellular motility.
			SOM.MK.1.BPM1.1.FTM.3.HCB.0737	Explain how a cell moves across a substrate by utilizing the concept of actin polymerization.
Lecture 2	HCB	Histology of Cellular Organelles	SOM.MK.1.BPM1.1.FTM.3.HCB.0738	Discuss the nucleus as a major organelle.
			SOM.MK.1.BPM1.1.FTM.3.HCB.0739	Give the function(s) of the nucleolus.
			SOM.MK.1.BPM1.1.FTM.3.HCB.0740	Describe the microscopic structure of the nucleolus.
			SOM.MK.1.BPM1.1.FTM.3.HCB.0741	State the three components of chromatin.
			SOM.MK.1.BPM1.1.FTM.3.HCB.0742	Describe how these three components come together to form chromatin.
			SOM.MK.1.BPM1.1.FTM.3.HCB.0743	Distinguish between euchromatin and heterochromatin in a nucleus.
			SOM.MK.1.BPM1.1.FTM.3.HCB.0744	Predict cellular activity based on the nuclear chromatin structure.
			SOM.MK.1.BPM1.1.FTM.3.HCB.0745	Give the function of the nuclear envelope.
			SOM.MK.1.BPM1.1.FTM.3.HCB.0746	Describe the structure of nuclear envelope including the nuclear pore complex.
			SOM.MK.1.BPM1.1.FTM.3.HCB.0747	Differentiate between the following organelles based on their microanatomical structures and function: ribosomes, endoplasmic reticulum (RER & SER), Golgi apparatus, lysosome, peroxisome and mitochondria.
			SOM.MK.1.BPM1.1.FTM.3.HCB.0748	Describe the bi-directional vesicle transport between the endoplasmic reticulum and Golgi apparatus.
			SOM.MK.1.BPM1.1.FTM.3.HCB.0749	Describe protein trafficking for lysosomes, secretion, plasma membrane, nucleus, mitochondria, cytosol and peroxisomes.
			SOM.MK.1.BPM1.1.FTM.3.HCB.0750	Describe the three cellular pathways to lysosomal degradation.
			SOM.MK.1.BPM1.1.FTM.3.HCB.0751	Identify the consequences of dysfunctional perixosomes for example Zellweger syndrome.
			SOM.MK.1.BPM1.1.FTM.3.HCB.0752	Identify the causes and consequences of lysosomal storage diseases (for example Tay-Sachs) based on the defective enzyme, accumulated products and organs most susceptible.
			SOM.MK.1.BPM1.1.FTM.3.HCB.0753	Define autophagy

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			SOM.MK.1.BPM1.1.FTM.3.HCB.0754	Describe the process of autophagy including the coordinated process from the membrane conformational changes to the fusion with lysosomes (autolysosomes).
			SOM.MK.1.BPM1.1.FTM.3.HCB.0755	Discuss the significance of autophagy
			SOM.MK.1.BPM1.1.FTM.3.HCB.0756	Give the structure and function (s) of the proteosome
			SOM.MK.1.BPM1.1.FTM.3.HCB.0757	Describe the structure of the proteasome.
			SOM.MK.1.BPM1.1.FTM.3.HCB.0758	Describe the five types of intracelluar inclusions.
			SOM.MK.1.BPM1.1.FTM.3.HCB.0759	Describe the four basic tissue types.
			SOM.MK.1.BPM1.1.FTM.3.HCB.0760	Identify the origin of the four basic types of tissue.
DLA 2	HCB	Cellular Biology of Vesicular Transport	SOM.MK.1.BPM1.1.FTM.3.HCB.0687	List the two major types of vesicular transport.
			SOM.MK.1.BPM1.1.FTM.3.HCB.0688	Describe the precise targeting of vesicles within the cell.
			SOM.MK.1.BPM1.1.FTM.3.HCB.0689	Describe the three different mechanisms of endocytosis.
			SOM.MK.1.BPM1.1.FTM.3.HCB.0690	Describe the role of endosomes in vesicular transport.
			SOM.MK.1.BPM1.1.FTM.3.HCB.0691	Describe the four pathways for processing internalized ligand-receptor complexes.
			SOM.MK.1.BPM1.1.FTM.3.HCB.0692	Describe the two general pathways of exocytosis.
DLA 3	BCHM	Structure and Function of Lipids	SOM.MK.I.BPM1.1.FTM.3.BCHM.0001	Review the grouping of lipids (Fatty acids, triacylglycerols, phospholipids, sphingolipids, steroids)
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0002	Review the significance and functions of cholesterol in humans.
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0003	Describe fatty acid structure. Discuss the relationship between chain length and desaturation and relate its significance to fluidity of cell membrane
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0004	Discuss importance of dietary essential fatty acids: linoleic acid (omega-6) and alpha-linolenic acid (omega-3)
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0005	Review the grouping of fatty acids into omega-6 and omega-3 families and describe the synthesis of arachidonic acid and of docosahexaenoic acid (DHA)
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0006	Describe structure-function relationship of triacylglycerols.
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0007	Distinguish between phospholipids (Sphingophospholipid and glycerophospholipids) and glycolipids with examples for each
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0008	Indicate composition and functions of: Glycerophospholipids (Phosphatidylcholine, phosphatidylethanolamine, phosphatidylserine, phosphatidylinositol, cardiolipin), Sphingophospholipid (Sphingomyelin), Glycolipids (Cerebrosides, globosides and gangliosides)
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0009	Discuss role of lung surfactant (Dipalmitoyl phosphatidylcholine/ DPPC) in respiratory distress syndrome
Lecture 3	PHYS	Homeostasis	SOM.MK.II.BPM1.1.FTM.3.PHYS.0001	Apply the concept of a homeostatic feedback loop to the regulation of key controlled variables (temperature, blood pressure, water content in extra- and intracellular compartments, pH) in the human body.
			SOM.MK.II.BPM1.1.FTM.3.PHYS.0002	List the typical values and normal ranges for plasma Na+, K+, H+ (pH), HCO3-, Cl-, Ca+2, and glucose, and the typical intracellular pH and concentrations of Na+, K+, Cl-, Ca+2, and HCO3.
			SOM.MK.II.BPM1.1.FTM.3.PHYS.0003	Define the terms "steady state" and "equilibrium" and evaluate if example systems like specific ion and water distributions across a membrane are in either state.
Lecture 4	PHYS	Tonicity and Osmolarity	SOM.MK.II.BPM1.1.FTM.3.PHYS.0007	Define Fick's Law of diffusion, and explain how changes in the concentration gradient, surface area, time and distance will influence the diffusional movement of a compound.
			SOM.MK.II.BPM1.1.FTM.3.PHYS.0020	Contrast the following units used to describe concentration: mM, mEq/l, mg/dl, mg%.
			SOM.MK.II.BPM1.1.FTM.3.PHYS.0021	List the typical values and normal range for plasma osmolality. Differentiate between the terms osmolarity and osmolality.
			SOM.MK.II.BPM1.1.FTM.3.PHYS.0017	Using a cell membrane as an example, define reflection coefficient and partition coefficient, and explain how the relative permeability of a cell to water and solutes will generate an osmotic pressure. Contrast the osmotic pressure generated across a cell membrane by a solution of particles that freely cross the membrane with that of a solution with the same osmolality, but particles that cannot cross the cell membrane.
			SOM.MK.II.BPM1.1.FTM.3.PHYS.0022	Describe the role of water channels (aquaporins) in facilitating the movement of water across biological membranes.
			SOM.MK.II.BPM1.1.FTM.3.PHYS.0023	Understand how regulation of the concentrations of K+, Cl-, and other Na+ solutes influence cell volume.
			SOM.MK.II.BPM1.1.FTM.3.PHYS.0024	Predict the movement of water across a semi-permeable membrane depending on solute concentrations in both compartments.
Lecture 5	PHYS	Resting Membrane Potential	SOM.MK.II.BPM1.1.FTM.3.PHYS.0025	Predict the difference in free energy of a solute and solvent between two compartments, including chemical and electrical forces. Evaluate if the compartments are in a steady state and/or equilibrium.
			SOM.MK.II.BPM1.1.FTM.3.PHYS.0026	Based on the principle of ionic attraction, explain how a potential difference across a membrane will influence the distribution of a cation and an anion.
			SOM.MK.II.BPM1.1.FTM.3.PHYS.0027	Write the Nernst equation and indicate how this equation accounts for both the chemical and electrical driving forces that act on an ion. Explain the effects of altering the intracellular or extracellular Na+, K+, Cl-, or Ca2+ concentration on the equilibrium potential for that ion.
			SOM.MK.II.BPM1.1.FTM.3.PHYS.0028	List the concentrations and equilibrium potentials of Na, K, Cl, Ca in a typical non-excitable cell. Based on the Nernst equilibrium potential, predict the direction of transmembrane flux for an ion when the membrane potential a) is at its equilibrium potential, b) is higher than the equilibrium potential, or c) is less than the equilibrium potential.
			SOM.MK.II.BPM1.1.FTM.3.PHYS.0029	Be able to calculate the equilibrium potential for an ion using the Nernst equation. Contrast the difference in the Nernst potential for K+ caused by a 5 mEq/l increase in extracellular K+ with the change in the Nernst potential for Na+ caused by a 5 mEq/l increase in extracellular Na+.

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			SOM.MK.II.BPM1.1.FTM.3.PHYS.0030	Using the Goldman-Hodgkin-Katz equation, explain how the relative permeabilities of the 4 key ion species (Na+, K+, Ca2+, Cl-) create the resting membrane potential. Given an increase or decrease in the permeability for each of the ions, predict how the membrane potential would change. (Not required to calculate on exams)
DLA 4	PHYS	Epithelial Function	SOM.MK.II.BPM1.1.FTM.3.PHYS.0014	Explain the functional significance of polarized distribution of ion channels and transporter proteins to the apical or basolateral cell membrane.
			SOM.MK.II.BPM1.1.FTM.3.PHYS.0018	Draw an epithelium, labeling the tight junctions, the apical membrane, and the basolateral membrane.
			SOM.MK.II.BPM1.1.FTM.3.PHYS.0349	Trace the movement of a solute that travels across an epithelium by a transcellular pathway and a solute that travels via a paracellular pathway.
			SOM.MK.II.BPM1.1.FTM.3.PHYS.0019	Explain the role of the tight junctions in delineating the apical and basolateral membranes.
Lecture 6	PHYS	Action Potentials	SOM.MK.II.BPM1.1.FTM.3.PHYS.0031	Define the following properties of ion channels: gating, activation, and inactivation.
			SOM.MK.II.BPM1.1.FTM.3.PHYS.0032	Contrast the gating of ion selective channels by extracellular ligands, intracellular ligands, stretch, and voltage.
			SOM.MK.II.BPM1.1.FTM.3.PHYS.0033	Know the properties of voltage-gated Na+, K+, and Ca2+ channels, and understand that voltage influences their gating, activation, and inactivation.
			SOM.MK.II.BPM1.1.FTM.3.PHYS.0034	Understand how the activity of voltage-gated Na+, K+, and Ca2+ channels generates an action potential and the roles of those channels in each phase (depolarization, overshoot, repolarization, hyperpolarization) of the action potential.
			SOM.MK.II.BPM1.1.FTM.3.PHYS.0035	Describe ionic basis of an action potential.
			SOM.MK.II.BPM1.1.FTM.3.PHYS.0036	Draw a typical action potential and explain why afterhyperpolarization coincides with the greatest fractional K conductance but not the greatest absolute K conductance.
			SOM.MK.II.BPM1.1.FTM.3.PHYS.0037	Using the Nernst equation and knowledge of sodium channel voltage dependent gating mechanisms (activation and inactivation gates), contrast the mechanisms leading to weakness and paralysis due to hypo- and hyper-kalemia.
			SOM.MK.II.BPM1.1.FTM.3.PHYS.0038	Draw a graph that shows how the voltage sensitivity of the voltage gated sodium channel changes with extracellular calcium concentration.
			SOM.MK.II.BPM1.1.FTM.3.PHYS.0039	Contrast the role that chronic depolarization (e.g. with hyperkalemia) plays in the activation of the K and Na conductances versus rapid depolarization (graded potential towards threshold).
DLA 5	ANAT	Early Embryology	SOM.MK.II.BPM1.1.FTM.1.ANAT.1001	Briefly describe the process from fertilization to the formation of the blastocyst, paying attention to the location and timeframe of each event
			SOM.MK.II.BPM1.1.FTM.1.ANAT.1002	Describe the location and process of implantation of the blastocyst
			SOM.MK.II.BPM1.1.FTM.1.ANAT.1003	List the derivatives of the germ layers.
			SOM.MK.II.BPM1.1.MSK.1.ANAT.1022	Describe the differentiation of the trophoblast, formation of lacunar networks and the establishment of a primordial uteroplacental circulation.
			SOM.MK.II.BPM1.1.FTM.1.ANAT.1023	Describe the sequence of events that leads to development of the amniotic cavity, bilaminar embryo, primitive umbilical vesicle (yolk sac), extraembryonic mesoderm and its clinical significance.
			SOM.MK.II.BPM1.1.FTM.1.ANAT.1024	Describe the development of the notochord and its role in induction of the neural plate.
			SOM.MK.II.BPM1.1.FTM.1.ANAT.1025	Describe the fate of the intraembryonic mesoderm, including the development of the somites and intraembryonic coelom.
			SOM.MK.II.BPM1.1.FTM.1.ANAT.1026	Describe the folding of the embryo, paying particular attention to the sequence and timing at which folding occurs.
			SOM.MK.II.BPM1.1.FTM.1.ANAT.1027	Describe the embryology and clinical presentation of sacrococcygeal teratoma
			SOM.MK.II.BPM1.1.FTM.1.ANAT.1028	Describe the process of gastrulation and its significance in embryonic development.
DLA 6	HCB	Histology of Basic Tissues - Epithelium	SOM.MK.I.BPM1.1.FTM.3.HCB.0767	Define epithelium.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0768	Describe the structural and functional characteristics that distinguish epithelial tissues from the three other basic tissues.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0769	Recap the localization of apical specializations (cilia, flagella, primary cilia, microvilli, stereocilia)
			SOM.MK.I.BPM1.1.FTM.3.HCB.0770	Recap the description of the terminal web.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0771	State the function of the glycocalyx.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0772	Describe the structure of the glycocalyx.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0773	Describe the classification of epithelial tissue.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0774	Differentiate between the different types of epithelia based on their structure, function and localization.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0775	Discuss the basic concepts of immotile cilia syndromes in Kartagener's syndrome.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0776	Explain the role of cell adhesion molecules (CAM) in cell adhesion
			SOM.MK.I.BPM1.1.FTM.3.HCB.0777	List the families of CAM in cell-to-cell adhesion (cadherins, selectins, immunoglobulin super family, integrins)
			SOM.MK.I.BPM1.1.FTM.3.HCB.0778	List the families of CAM in cell-to-ECM adhesion (integrins)
			SOM.MK.I.BPM1.1.FTM.3.HCB.0779	State the function and localization of zonula occludens.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0780	Describe the histological structure of zonula occludens.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0781	State the function and localization of zonula adherens.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0782	Describe the histological structure of zonula adherens.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0783	State the function and localization of gap junctions.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0784	Describe the histological structure of gap junctions.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0785	State the function and localization of desmosomes.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0786	Describe the histological structure of desmosomes.

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Lecture 7	HCB	Histology of Epithelium & Glands	SOM.MK.I.BPM1.1.FTM.3.HCB.0787	Describe the terminal bar.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0788	State the function and localization of hemidesmosomes.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0789	Describe the histological structure of hemidesmosomes.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0790	State the function and localization of focal adhesions.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0791	Describe the histological structure of focal adhesions.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0792	Give function(s) of lateral digitations.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0793	Describe the structure of lateral digitations.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0794	Identify lateral digitations on micrographs.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0795	Give function(s) of basal infoldings.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0796	Describe the structure of basal infoldings.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0797	Identify basal infoldings on micrographs.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0798	Give the function of the basal lamina.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0799	Describe the microscopic structure of the basal lamina.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0800	Identify the basal lamina on micrographs.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0801	Describe the microscopic structure of the basement membrane.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0802	Compare and contrast basement membrane vs. basal lamina vs external lamina.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0803	Compare and contrast exocrine vs. endocrine glands.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0804	Distinguish between unicellular and multicellular exocrine glands.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0805	Describe the three modes of secretion in multicellular exocrine glands.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0806	Describe the classification of multicellular exocrine glands.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0807	Describe the microscopic appearance of serous glands.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0808	Describe the microscopic appearance of mucous glands.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0809	Describe the microscopic appearance of mixed glands.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0810	Describe the histology of a compound salivary gland.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0811	Identify parotid, submandibular and sublingual salivary glands on photomicrographs.
DLA 7	BCHM	Structure and Function of Nucleic Acids	SOM.MK.I.BPM1.1.FTM.3.BCHM.0020	Distinguish between purines, pyrimidines; ribonucleosides, deoxyribonucleosides; ribonucleotides and deoxyribonucleotides.
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0021	Describe the phosphodiester bond
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0022	Describe the three important features of the DNA double helix (complementary, antiparallel, and forces stabilizing the helix).
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0023	Discuss the physicochemical properties of DNA
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0024	Describe the basic structures and roles of the major classes of RNA (mRNA, tRNA, rRNA, snRNA, miRNA)
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0025	Predict the sequence of the complementary strand if the sequence of one of the DNA strands is provided.
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0026	Analyze the nucleotide composition of double stranded DNA using the Chargaff's rule
DLA 8	BCHM	DNA Packaging	SOM.MK.I.BPM1.1.FTM.3.BCHM.0028	Differentiate between a supercoiled and a relaxed DNA molecule
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0029	State the reasons why supercoiling of DNA molecules is important.
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0030	Compare and contrast DNA packaging in prokaryotes and eukaryotes.
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0031	Describe the structure of chromatin and histone proteins.
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0032	Explain how dsDNA and histones associate to form the nucleosome.
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0033	Describe the different structural conformations of DNA packaging in eukaryotes (10 nm Fiber , 30 nm fiber and metaphase chromosomes).
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0034	Describe how histone modification can influence transcriptional activity
DLA 9	BCHM	DNA Replication and Telomerase	SOM.MK.I.BPM1.1.FTM.3.BCHM.0048	Describe the role of telomeres and telomerase in eukaryotic DNA replication
Lecture 8	BCHM	DNA Replication	SOM.MK.I.BPM1.1.FTM.3.BCHM.0035	Describe semi-conservation replication of DNA
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0036	Compare and contrast the similarities and differences between DNA Pol I and DNA Pol III of E. coli
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0037	Describe how DNA is synthesized from its 5' to 3' end
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0038	Explain the significance of the origin of replication during the process of DNA synthesis.
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0039	Explain the problems associated with unwinding of the DNA double helix
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0040	Describe the role of helicase enzymes in unwinding DNA prior to its replication
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0041	Describe the role of single-strand binding protein & topoisomerases in DNA replication
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0042	Outline the order of events from RNA priming to completed DNA during replication in E.coli.
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0043	Distinguish between synthesis of leading and lagging strands of DNA. Identify the significance of DNA ligase in lagging strand synthesis
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0044	Describe the significance of the various proteins and enzymes required for replication and predict the effects of mutations of these proteins and enzymes on DNA replication
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0045	Evaluate the proofreading activity of DNA polymerase
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0046	Differentiate between DNA replication in eukaryotes and prokaryotes
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0047	Outline the mechanisms of action of drugs that interfere with DNA replication
DLA 10	BCHM	RNA Polymerase and Transcription	SOM.MK.I.BPM1.1.FTM.3.BCHM.0049	Describe the structural features of a gene and their functions
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0050	Describe the types and functions of the different cellular RNAs (mRNA, tRNA, rRNA, snRNA, miRNA)
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0051	Describe the major steps of RNA synthesis in prokaryotes and eukaryotes
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0052	Explain the function and structure of DNA sequences that are involved in the initiation (promoters) and termination of transcription
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0053	Predict the sequence of the transcribed RNA if the template DNA strand or the coding strand of the RNA is provided
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0054	List the RNA products that are transcribed by the three different types of RNA polymerases in eukaryotes
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0057	Explain the mechanism of action of rifampin and alpha-amanitin as inhibitors of transcription.

Activity	Discipline	Title	OBJECTIVE CODE	Lecture Objective
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0058	Describe the effect of the antibiotic drugs that interfere with DNA gyrase such as (quinolones) ciprofloxacin
Lecture 9	BCHM	Post-transcriptional Modifications	SOM.MK.I.BPM1.1.FTM.3.BCHM.0059	Differentiate between the structures of prokaryotic and eukaryotic mRNAs
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0060	Describe the post-transcriptional modifications of eukaryote mRNA and the significance of these modifications (5' capping, polyadenylation, and splicing)
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0061	Predict effects of splice site mutations on mature mRNA and ultimately the translated protein
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0062	Describe how RNA editing affects our understanding of transmission of genetic information from the genome to protein
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0063	Describe human disease conditions that may be associated with mRNA modification and errors in this process
Lecture 10	GNET	Mitosis, Meiosis and Non-disjunction	SOM.MKIII.BPM1.1.FTM.3.GNET.0001	Recognize differences in rates of cell division in different tissues
			SOM.MKIII.BPM1.1.FTM.3.GNET.0002	Analyze stages of the cell cycle - (G1, G0, S, G2, M)
			SOM.MKIII.BPM1.1.FTM.3.GNET.0003	Indicate the role of p53 in the cell cycle: Role in genome surveillance and its role in initiation of apoptosis
			SOM.MKIII.BPM1.1.FTM.3.GNET.0004	Describe the chromosomal structure changes in different phases of the cell cycle
			SOM.MKIII.BPM1.1.FTM.3.GNET.0005	Explain the significance of mitosis, summarize the events, and analyze structures of mitosis including Prophase: Events in the nucleus, formation of spindle in cytoplasm, Metaphase: Association of chromosomes with spindle fibres, Anaphase: Separation of sister chromatids, Telophase: Cytokinesis, re-formation of nuclei
			SOM.MKIII.BPM1.1.FTM.3.GNET.0006	Recognize the significance of meiosis and summarize its main achievements
			SOM.MKIII.BPM1.1.FTM.3.GNET.0007	Describe the sequence of events that comprise meiosis.
			SOM.MKIII.BPM1.1.FTM.3.GNET.0008	Analyze the effect of a nondisjunction of the sex chromosomes in meiosis 1 or 2 during gametogenesis which lead to: Turner, Klinefelter, XYY, or XXX syndromes.
			SOM.MKIII.BPM1.1.FTM.3.GNET.0009	Interpret the significance of recombination in prophase I of meiosis. Identify the role of meiosis in gametogenesis. Analyze events and structures that occur in meiosis
			SOM.MKIII.BPM1.1.FTM.3.GNET.0010	Compare and contrast the overall features of mitosis with meiosis and explain how meiosis introduces diversity into the genotype of the offsping after sexual reproduction
			SOM.MKIII.BPM1.1.FTM.3.GNET.0011	Describe the distinguishing features and the morphological classification of chromosomes including metacentric, submetacentric, acrocentric, and telocentric
			SOM.MKIII.BPM1.1.FTM.3.GNET.0012	Outline the differences between meiosis in males and females
			SOM.MKIII.BPM1.1.FTM.3.GNET.0013	Compare and contrast the outcome of nondisjunction that might occur during meiosis or in mitosis
Lecture 11	GNET	Genomics and Human Variation	SOM.MKIII.BPM1.1.FTM.3.GNET.0016	Compare and contrast different types of repetitive DNA including low copy repeats, SSR, VNTR, LINE, SINE, pseudogenes, and gene families
			SOM.MKIII.BPM1.1.FTM.3.GNET.0017	Compare and contrast single nucleotide polymorphisms to rare variants
			SOM.MKIII.BPM1.1.FTM.3.GNET.0018	Discuss the structure of chromosomes in the human (male and female), and general terminology used to describe karyotypes; Compare nuclear DNA to organellar DNA (such as the mitochondrial
			SOM.MKIII.BPM1.1.FTM.3.GNET.0019	Analyze dosage compensation by X-chromosome inactivation
			SOM.MKIII.BPM1.1.FTM.3.GNET.0105	Explain the normal function of epigenetic gene silencing in development; including the role of X-inactivation as discussed Genomics and Human Variation
DLA 11	GNET	Inheritance: AD and AR and Pseudo Dominance	SOM.MKIII.BPM1.1.FTM.3.GNET.0020	Describe basic definitions of human genetics including: locus, gene, allele, genotype, and zygosity
			SOM.MKIII.BPM1.1.FTM.3.GNET.0021	From a description of a pedigree, identify autosomal dominant (AD) and autosomal recessive (AR) inheritance; Compare and contrast Mendelian and non-Mendelian genetics
			SOM.MKIII.BPM1.1.FTM.3.GNET.0022	Calculate recurrence risk of autosomal dominant and autosomal recessive disorders within families
			SOM.MKIII.BPM1.1.FTM.3.GNET.0023	Analyze family pedigree to determine the risk that immediate family members might be carriers of disease alleles (AR or X-linked) and when to use the 2/3 rule
			SOM.MKIII.BPM1.1.FTM.3.GNET.0024	Describe and differentiate between pseudoatosomal dominance, co-dominance, and incomplete dominance; give examples
			SOM.MKIII.BPM1.1.FTM.3.GNET.0025	Describe co-dominance, giving examples
			SOM.MKIII.BPM1.1.FTM.3.GNET.0026	Describe incomplete dominance, giving examples
Lecture 12	GNET	Flipped Class: Mendelian Inheritance: AD and AR	SOM.MKIII.BPM1.1.FTM.3.GNET.0027	Interpret a family pedigree incorporating standard genetic symbols. Interpret a pedigree and identify features, disorders, genotype, carrier status, recurrence risk, and genetic principles
			SOM.MKIII.BPM1.1.FTM.3.GNET.0028	Demonstrate how disease liability is transmitted for autosomal dominant and autosomal recessive disorders, beginning from the first "de novo" occurance
			SOM.MKIII.BPM1.1.FTM.3.GNET.0029	Propose how the basic risk calculations are performed for the different categories of genetic transmission: AR and AD
			SOM.MKIII.BPM1.1.FTM.3.GNET.0030	Identify obligate carriers in a pedigree: AR, or AD with reduced penetrance
			SOM.MKIII.BPM1.1.FTM.3.GNET.0031	Distinguish the mode of inheritance, identify the main clinical features, and interpret genetic principles illustrated by the following autosomal dominant disorders: Huntington disease, myotonic dystrophy, familial hypercholesterolemia, osteogenesis imperfecta, Marfan syndrome, Achondroplasia, NF, acute intermittent porphyria)

Activity	Discipline	Title	OBJECTIVE CODE	Lecture Objective
		Inheritance: AD and AR	SOM.MKIII.BPM1.1.FTM.3.GNET.0032	Distinguish the mode of inheritance, identify the main clinical features, and interpret genetic principles illustrated by the following following autosomal recessive disorders: CF, PKU, Tay-Sachs disease, Sickle cell disease, thalassemia, hemochromatosis, homocystineuria, SCID due to ADA deficiency, alkaptonuria
			SOM.MKIII.BPM1.1.FTM.3.GNET.0033	Identify pseudodominant inheritance and indicate factors that result in this inheritance pattern. Be able to identify this concept from clinical scenarios and pedigrees
			SOM.MKIII.BPM1.1.FTM.3.GNET.0034	Compare and contrast locus heterogeneity and allelic heterogeneity; give examples
			SOM.MKIII.BPM1.1.FTM.3.GNET.0035	Explain how allelic heterogeneity allows the formation of a compound heterozygote in autosomal recessive disorders
			SOM.MKIII.BPM1.1.FTM.3.GNET.0036	Define the terms and give examples for: haploinsufficiency, dominant negative effect, gain of function mutations, loss of function mutations
DLA 12	GNET	Inheritance: X-linkage	SOM.MKIII.BPM1.1.FTM.3.GNET.0037	Distinguish the mode of inheritance, identify the main clinical features, and interpret genetic principles illustrated by the following X-linked disorders: Hemophilia A and B, G6PD deficiency, Duchenne and Becker muscular dystrophy, red/green color blindness. Lesch Nyhan syndrome, X-SCID as examples of X-linked disorders
			SOM.MKIII.BPM1.1.FTM.3.GNET.0038	Summarize and distinguish the characteristics of the following patterns of inheritance using pedigrees: autosomal dominant, autosomal recessive, X-linked dominant and X-linked recessive
			SOM.MKIII.BPM1.1.FTM.3.GNET.0039	Demonstrate how disease liability is transmitted for X-linked diseases
			SOM.MKIII.BPM1.1.FTM.3.GNET.0040	Propose how the basic risk calculations are performed for X-linked disorders
			SOM.MKIII.BPM1.1.FTM.3.GNET.0041	Explain how a woman who is a carrier of an X-linked disorder might manifest symptoms of the condition (manifesting heterozygote) Skewed X-inactivation, non-random X–inactivation
			SOM.MKIII.BPM1.1.FTM.3.GNET.0042	Discuss X-linked dominant disorders and traits
			SOM.MKIII.BPM1.1.FTM.3.GNET.0043	Compare and contrast variable expressivity and incomplete penetrance; Give examples of each
			SOM.MKIII.BPM1.1.FTM.3.GNET.0044	Describe pleiotropy
			SOM.MKIII.BPM1.1.FTM.3.GNET.0045	Compare and contrast locus heterogeneity and allelic heterogeneity; gve examples
			SOM.MKIII.BPM1.1.FTM.3.GNET.0046	Describe locus heterogeneity in terms of complementation groups
			SOM.MKIII.BPM1.1.FTM.3.GNET.0047	Explain how allelic heterogeneity allows the formation of a compound heterozygote in autosomal recessive disorders
Lecture 13	GNET	Flipped Class: X-Linked Inheritance	SOM.MKIII.BPM1.1.FTM.3.GNET.0048	Demonstrate how disease liability is transmitted for autosomal dominant diseases, autosomal recessive diseases and X-linked diseases
			SOM.MKIII.BPM1.1.FTM.3.GNET.0049	Distinguish the different characteristics of recessive and dominant X-linked traits using pedigrees
			SOM.MKIII.BPM1.1.FTM.3.GNET.0050	Identify obligate carriers in a pedigree depicting an X-linked trait
			SOM.MKIII.BPM1.1.FTM.3.GNET.0051	Compare and contrast between X-linked, Y-linked, AR, AD, and mitochondrial traits.
			SOM.MKIII.BPM1.1.FTM.3.GNET.0052	Discuss the concept of penetrance and variable expression and Differentiate between locus and allelic heterogeneity
			SOM.MKIII.BPM1.1.FTM.3.GNET.0053	Calculate the recurrence risk based on penetrance for autosomal dominant and autosomal recessive disorders
			SOM.MKIII.BPM1.1.FTM.3.GNET.0054	Define pleiotropy with examples. Be able to identify concepts if given a clinical scenario
			SOM.MKIII.BPM1.1.FTM.3.GNET.0055	Recognize occurrence of new mutations as a cause of new genetic diseases in families. Be able to identify this concept from clinical scenarios and pedigrees
			SOM.MKIII.BPM1.1.FTM.3.GNET.0056	Explain the phenomenon of germinal (germ-line) mosaicism
			SOM.MKIII.BPM1.1.FTM.3.GNET.0057	Identify genetic disorders that have a delayed age of onset, and be able to identify this concept from clinical scenarios and pedigrees
			SOM.MKIII.BPM1.1.FTM.3.GNET.0058	Explain the phenomenon of a manifesting heterozygote in X-linked recessive disorders, and be able to identify this concept from clinical scenarios and pedigrees
DLA 13	BCHM	Genetic code and Translation	SOM.MK.I.BPM1.1.FTM.3.BCHM.0064	Explain why the genetic code is collinear with DNA, triplet in nature, redundant (degenerate), and non-overlapping.
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0065	Define the terms: reading frame and frame-shift mutation.
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0066	Interpret the standard genetic code table. Identify the initiation codon and the three termination codons.
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0067	Explain the universal nature of the genetic code (with minor differences in mitochondria) and predict the sequence of a peptide if the coding strand of the DNA is provided.
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0068	Discuss how protein synthesis proceeds from the N-terminal to their C-terminal ends and explain how this creates a naming convention for peptides.
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0050	List and describe the major function of the following types of RNA in cells: mRNA, tRNA, rRNA, snRNA, miRNA.
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0070	Describe the subunit composition prokaryotic (70S) and eukaryotic (80S) ribosomes. Recognize the roles of A, P, and E sites on the ribosome.
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0071	Describe the structure, function, and charging of tRNAs.
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0072	Describe the codon/anticodon interaction and discuss the wobble hypothesis.
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0073	Describe the sequence of events that occurs during translation in prokaryotes (initiation, elongation and termination) and list the major differences between prokaryotic and eukaryotic translation.
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0074	Explain how miRNA and diphtheria toxin interfere with eukaryotic translation.

Activity	Discipline	Title	OBJECTIVE CODE	Lecture Objective
Lecture 14	BCHM	Translation and Post-translational Modification	SOM.MK.I.BPM1.1.FTM.3.BCHM.0069	Explain the mode of action of common antibiotics that interfere with translation: Initiation inhibitors (streptomycin), Elongation inhibitors (Tetracycline, chloramphenicol, erythromycin, puromycin, cycloheximide).
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0075	Review examples of post-translational modifications: Zymogen activation (trypsinogen to trypsin), Serine/threonine phosphorylation (regulation of enzymes in metabolism), Tyrosine phosphorylation (Insulin receptor), O-linked glycosylation, N-linked glycosylation and Lipid anchoring (farnesyl groups to Ras).
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0076	Describe proteolytic processing and post-translational modifications using insulin as an example.
DLA 14	BCHM	Amino Acids: Structure & Function	SOM.MK.I.BPM1.1.FTM.3.BCHM.0078	Classify amino acids based on their R groups (with examples for each group).
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0079	Name the amino acids with a charged side chain, the amino acids containing a hydroxyl group and the amino acids containing sulphur.
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0081	Discuss the role of histidine residues in buffering in hemoglobin.
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0083	Discuss the importance of derivatives of amino acids (GABA, histamine, serotonin, nitric oxide, and catecholamines).
Lecture 15	BCHM	Structure and Function of Proteins	SOM.MK.I.BPM1.1.FTM.3.BCHM.0087	Differentiate between the hierarchical levels of protein structure using the example of myoglobin and hemoglobin.
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0088	Discuss the significance of bond forces involved in maintenance of protein structure (peptide bonds, disulfide bonds, hydrogen bonds, ionic bonds, hydrophobic forces and Van der Waals forces) using hemoglobin as an example.
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0090	Relate that missense mutations can alter the solubility, function, charge and electrophoretic properties of proteins.
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0091	Discuss neurochemical abnormalities in Prion disease and Alzheimer disease.
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0092	Describe the role of chaperones and ubiquitin.
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0093	Discuss protein denaturation in vivo and in the laboratory.
Lecture 16	GNET	Unexpected Events in Inheritance	SOM.MKIII.BPM1.1.FTM.3.GNET.0059	Explain how the classic pattern of inheritance may be disrupted by incomplete penetrance, variable expressivity and variable age-of-onset.
			SOM.MKIII.BPM1.1.FTM.3.GNET.0060	Describe examples of genetic abnormalities that may result in a non-Mendelian pattern of inheritance
			SOM.MKIII.BPM1.1.FTM.3.GNET.0061	Discuss disorders due to mutations in mitochondrial genes: Leber hereditary optic neuropathy (LHON), MELAS, MERRF
			SOM.MKIII.BPM1.1.FTM.3.GNET.0062	Identify the phenomenon of heteroplasmy in mitochondrial disorders and indicate its genetic basis
			SOM.MKIII.BPM1.1.FTM.3.GNET.0063	Predict the effect of imprinting in the transmission of genetic disease
			SOM.MKIII.BPM1.1.FTM.3.GNET.0064	Identify imprinting defects (Prader Willi syndrome, Angelman syndrome)
			SOM.MKIII.BPM1.1.FTM.3.GNET.0065	Explain the phenomenon of uniparental disomy and microdeletion in Prader-Willi and Angelman syndrome
			SOM.MKIII.BPM1.1.FTM.3.GNET.0066	Analyze how triplet repeat expansion mutations result in anticipation (non-Mendelian inheritance) for Huntington disease, myotonic dystrophy, Fragile X syndrome.
			SOM.MKIII.BPM1.1.FTM.3.GNET.0067	Explain digenic inheritance using retinitis pigmentosa as an example
			SOM.MKIII.BPM1.1.FTM.3.GNET.0068	Determine a strategy for identifying Non-Mendelian disorders, investigating them and providing appropriate genetic counselling for families
			SOM.MKIII.BPM1.1.FTM.3.GNET.0110	Demonstrate how genomic imprinting would appear in a human pedigree when either a male or female passes a mutation to offspring
Reading Assignment	GNET	Molecular Biology Techniques	SOM.MK.I.BPM1.1.FTM.3.GNET.0200	Explain the general principles and utilization of various molecular biology tools and techniques, including Restriction endonucleases, DNA cloning, Probes, Southern Blotting, Restriction Fragment Length Polymorphisms, Polymerase chain reaction and Gene Expression analysis
			SOM.MK.I.BPM1.1.FTM.3.GNET.0201	Summarize how complementary DNA (cDNA) is generated from messenger RNA (mRNA) and how this technique is used to express human proteins in a bacteria to generate therapeutic proteins such as insulin
Lecture 17	GNET	Molecular Biology Techniques	SOM.MK.I.BPM1.1.FTM.3.GNET.0202	Identify the differences between mutations which affect the structures of a gene (or gene region), the structure of the transcribed mRNA product and the structure of the translated protein
			SOM.MK.I.BPM1.1.FTM.3.GNET.0203	Explain the concept of hybridization: between a complementary single stranded DNA probe and its target (DNA or RNA); between an antibody and its protein target
			SOM.MK.I.BPM1.1.FTM.3.GNET.0204	Compare data from RFLP, ASO, DNA sequencing: from normal homozygotes, carriers (heterozygotes), and affected homozygotes
			SOM.MK.I.BPM1.1.FTM.3.GNET.0205	Predict the application of molecular biology techniques for detection of a single base mutation versus small insertion or deletion of DNA sequence, versus large insertion or deletion of DNA sequence.
Reading Assignment	BCHM	Properties of Enzymes	SOM.MK.I.BPM1.1.FTM.3.BCHM.0103	Describe the factors that influence enzyme reaction rates (substrate, pH and temperature).
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0104	Discuss enzyme regulation via the concentrations of substrates or products.
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0105	Describe enzyme regulation via the modulation of enzyme concentrations.
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0106	Discuss conformation of enzymes and changes of conformation in relation to enzyme regulation. (reversible covalent modification, proteolytic cleavage and allosteric regulation).
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0107	Describe the concept of proteolytic activation of digestive enzymes (eg: zymogens such as trypsinogen).
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0108	Discuss the kinetics of allosteric enzymes and explain the sigmoidal graph.

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Lecture 18	BCHM	Enzymes	SOM.MK.I.BPM1.1.FTM.3.BCHM.0109	Describe allosteric enzymes and their regulation related to changes of K0.5 or Vmax. Discuss in general the regulation of metabolic pathways related to feedback inhibition and feed-forward activation.
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0110	Describe Michaelis-Menten kinetics
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0111	Define Km and Vmax and discuss their significance with the help of a graph.
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0112	Discuss the inhibition caused by statin drugs (enzyme inhibited, effect on kinetics).
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0113	Compare and contrast the mode of action and kinetics of reversible competitive and noncompetitive inhibitors.
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0114	Compare and contrast the Michelis-Menten and Lineweaver-Burk plots.
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0115	Indicate X- and Y- intercepts in the Lineweaver-Burk plot
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0116	Compare and contrast competitive and noncompetitive inhibition using Michaelis-Menten graph and Lineweaver-Burk plot.
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0117	Distinguish between reversible and irreversible inhibitors.
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0118	Discuss inhibition caused by DFP and aspirin (enzymes inhibited, effects of inhibition).
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0119	Discuss suicide inhibition of enzymes using allopurinol as example.
DLA 15	GNET	Genomics in Diagnosis	SOM.MKIII.BPM1.1.FTM.3.GNET.0069	Explain how triplet repeat expansion disorders are diagnosed
			SOM.MKIII.BPM1.1.FTM.3.GNET.0070	Discuss why each triplet repeat expansion disorder needs its own unique test for diagnosis
			SOM.MKIII.BPM1.1.FTM.3.GNET.0071	Explain how a G-band karyotype is prepared
			SOM.MKIII.BPM1.1.FTM.3.GNET.0072	Describe FISH and how it can be used in diagnostic applications
			SOM.MKIII.BPM1.1.FTM.3.GNET.0073	Compare and contrast : Metaphase FISH, Interphase FISH, and Chromosome painting (SKY FISH, or spectral karyotype)
			SOM.MKIII.BPM1.1.FTM.3.GNET.0074	Describe the strategy behind the microarray: CGH Microarray, SNP Microarray, cDNA Microarray
Lecture 19	GNET	Use of Genomics in Diagnosis	SOM.MKIII.BPM1.1.FTM.3.GNET.0075	Analyze genomic approaches that are used in the diagnosis of human disorders
			SOM.MKIII.BPM1.1.FTM.3.GNET.0076	Analyze how PCR can be used to obtain a diagnosis of a genetic disorder
			SOM.MKIII.BPM1.1.FTM.3.GNET.0077	Discuss how next generation sequencing is being used in genomic applications
			SOM.MKIII.BPM1.1.FTM.3.GNET.0078	Describe the difference between whole genome sequencing (WGS), whole exome sequencing (WES), gene panels, and single targeted gene sequencing
			SOM.MKIII.BPM1.1.FTM.3.GNET.0079	Describe the different types of mutations that may be observed in the human genome and explain how they are categorized.
			SOM.MKIII.BPM1.1.FTM.3.GNET.0080	Consider the relative pros and cons of the genomic and genetic techniques described in this section, and their limits of resolution
Lecture 20	BCHM	Signal Transduction	SOM.MK.I.BPM1.1.FTM.3.BCHM.0120	Describe different types of cell signalling: endocrine, autocrine, paracrine, neuronal & contact-dependent.
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0121	Indicate four different classes of receptors (steroid receptors, ion-channel receptor, receptor enzyme, receptors that form second messengers including G-protein-coupled receptors) with specific examples for each class.
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0122	Discuss events involved in signal transduction
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0123	Identify the location of the receptors for steroid hormones. Describe intracellular signal transduction by steroid hormones.
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0124	Describe the molecular changes when insulin binds to its receptor.
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0125	Describe the different target (effector) enzymes of G proteins.
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0126	Describe the two major second messenger systems (adenylate cyclase and phosphoinositide systems) associated with G-proteins and signal termination.
Lecture 21	BCHM	Second Messenger Systems	SOM.MK.I.BPM1.1.FTM.3.BCHM.0127	Explain how cholera toxin modifies Gs and how pertussis toxin modifies Gi. Predict the effects of these modifications in the respective cells.
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0128	Discuss the roles of cAMP, IP3, DAG and Calcium as mediators of signal transduction in different cell types.
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0129	Review the mechanism of action of phosphodiesterase inhibitors.
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0130	Describe activation of adenylate cyclase by Gs alpha and phospholipase C by Gq-alpha subunits.
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0131	Describe the formation and biological role of cGMP. Indicate significance of guanylate cyclase.
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0132	Identify mechanism of action of nitric oxide (nitroglycerin) and its role in smooth muscle relaxation and vasodilatation.
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0133	Compare the cellular activation of protein kinases A, C and G.
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0134	Distinguish signaling mechanisms associated with alpha-1, alpha-2 and beta-adrenergic receptors, glucagon receptors and insulin receptors.
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0135	Predict the changes when an agonist or antagonist of a specific signaling pathway is administered.
Lecture 22	GNET	Flipped Class: Genetics and Genomics Cases	SOM.MKIII.BPM1.1.FTM.3.GNET.0081	Analyze examples of molecular diagnosis obtained from genetic and genomic laboratory analysis
Lecture 23	BCHM	Clinical Enzymology	SOM.MK.I.BPM1.1.FTM.3.BCHM.0136	Review the significance of estimation of LDH.
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0137	Differentiate isozymes of creatine kinase based on tissue location and subunit composition.
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0138	Discuss utility of serum injury markers following myocardial infarction: Myoglobin, CK-MB/total CK ratio, serum cardiac troponins I and T
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0139	Discuss the utility of serum injury markers following MI: Time frame of serum injury markers (myoglobin, cardiac troponins, CK-MB) with the help of a graph.
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0140	Describe the application of ALT, AST, ALP and GGT as markers of hepatocellular disease, biliary disease and alcohol liver disease.
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0141	Discuss significance of serum amylase and serum lipase related to pancreatic disease.

Activity	Discipline	Title	OBJECTIVE CODE	Lecture Objective
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0142	Indicate specific serum injury markers for bone disease (ALP), prostate cancer (PSA) and liver cancer (alpha-fetoprotein).
Lecture 24	HCB	Apoptosis and Necrosis	SOM.MK.I.BPM1.1.FTM.3.HCB.0819	Define apoptosis
			SOM.MK.I.BPM1.1.FTM.3.HCB.0820	Define necrosis
			SOM.MK.I.BPM1.1.FTM.3.HCB.0821	State the importance of apoptosis
			SOM.MK.I.BPM1.1.FTM.3.HCB.0822	State the importance of necrosis
			SOM.MK.I.BPM1.1.FTM.3.HCB.0823	Discuss common causes of cellular injury.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0824	Compare reversible vs. irreversible injuries.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0825	Describe the early cellular responses to injury.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0826	Discuss some examples of physiological processes that lead to cell death by apoptosis
			SOM.MK.I.BPM1.1.FTM.3.HCB.0827	Discuss some examples of pathological processes that lead to cell death by necrosis
			SOM.MK.I.BPM1.1.FTM.3.HCB.0828	Discuss the morphological changes associated with cell death by apoptosis.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0829	Discuss the morphological changes associated with cell death by necrosis.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0830	Discuss the biochemical features associated with cell death by apoptosis.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0831	Discuss the biochemical features associated with cell death by necrosis.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0832	Describe the intrinsic apoptosis pathway.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0833	Describe the extrinsic apoptosis pathway.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0834	Describe the role of caspases in apoptosis.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0835	Discuss methods to assess apoptosis.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0836	Discuss methods to assess necrosis.
Lecture 25	HCB	Cellular Adaptation	SOM.MK.I.BPM1.1.FTM.3.HCB.0837	Define the term "hypertrophy" and list examples.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0838	Define the term "hyperplasia" and list examples.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0839	Define the term "atrophy" and list examples.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0840	Define the term "involution" and list example(s).
			SOM.MK.I.BPM1.1.FTM.3.HCB.0841	Define the term "metaplasia" and list examples.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0842	Define the terms "failure of differentiation" and "cellular atypia".
			SOM.MK.I.BPM1.1.FTM.3.HCB.0843	Define the term "dysplasia" and list examples.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0844	Define the term "neoplasia" and list examples.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0845	Compare and contrast benign vs. malignant neoplasms.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0846	Define the terms "carcinoma in situ" and "invasion".
			SOM.MK.I.BPM1.1.FTM.3.HCB.0847	Describe the classification of cancer cells including carcinoma, sarcoma and leukemia.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0848	Discuss the hallmarks of cancer.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0849	Describe tumor growth and progression.
			SOM.MK.I.BPM1.1.FTM.3.HCB.0850	Define the term "primary tumor".
			SOM.MK.I.BPM1.1.FTM.3.HCB.0851	Define the term "metastases" (secondary tumors).
			SOM.MK.I.BPM1.1.FTM.3.HCB.0852	List the main modes for spread of malignant neoplasms.
Lecture 26	BCHM	Molecular Mechanisms in Inherited Disorders	SOM.MK.I.BPM1.1.FTM.3.BCHM.0179	Describe the molecular basis for clinical features, diagnostic tests and management of Cystic Fibrosis
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0180	Describe the molecular basis for clinical features, diagnostic tests and management of Sickle cell anemia
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0181	Describe the molecular basis for clinical features and diagnostic tests in Duchenne Muscular Dystrophy and Becker Muscular Dystrophy
			SOM.MK.I.BPM1.1.FTM.3.BCHM.0182	Compare the differences between Duchenne muscular dystrophy and Becker muscular dystrophy with reference to severity of the disorder, type of mutations, immunohistochemistry findings and alterations in dystrophin